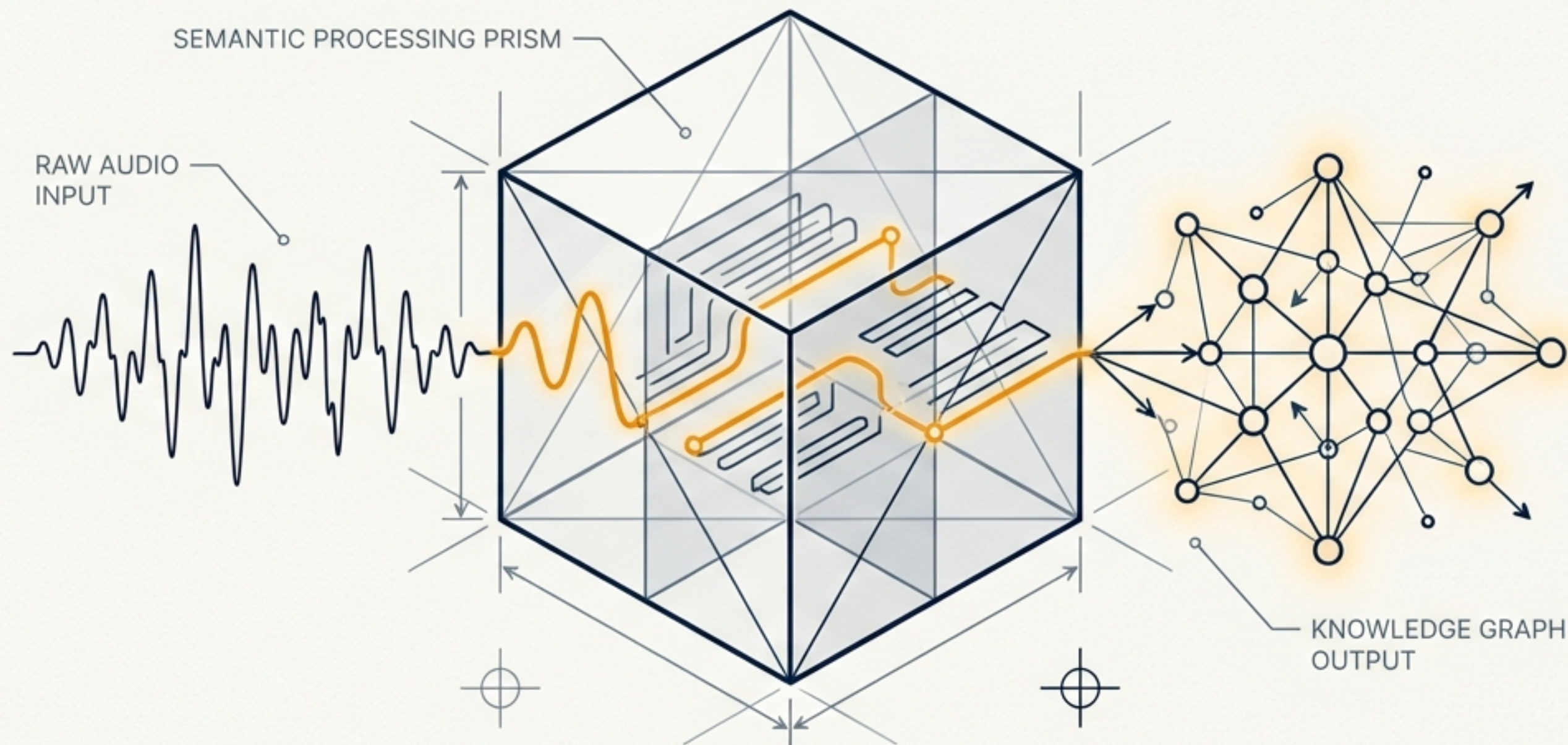


THE GLASS BOX ARCHITECTURE

Building a Semantic Graph-Powered Podcast Recommender



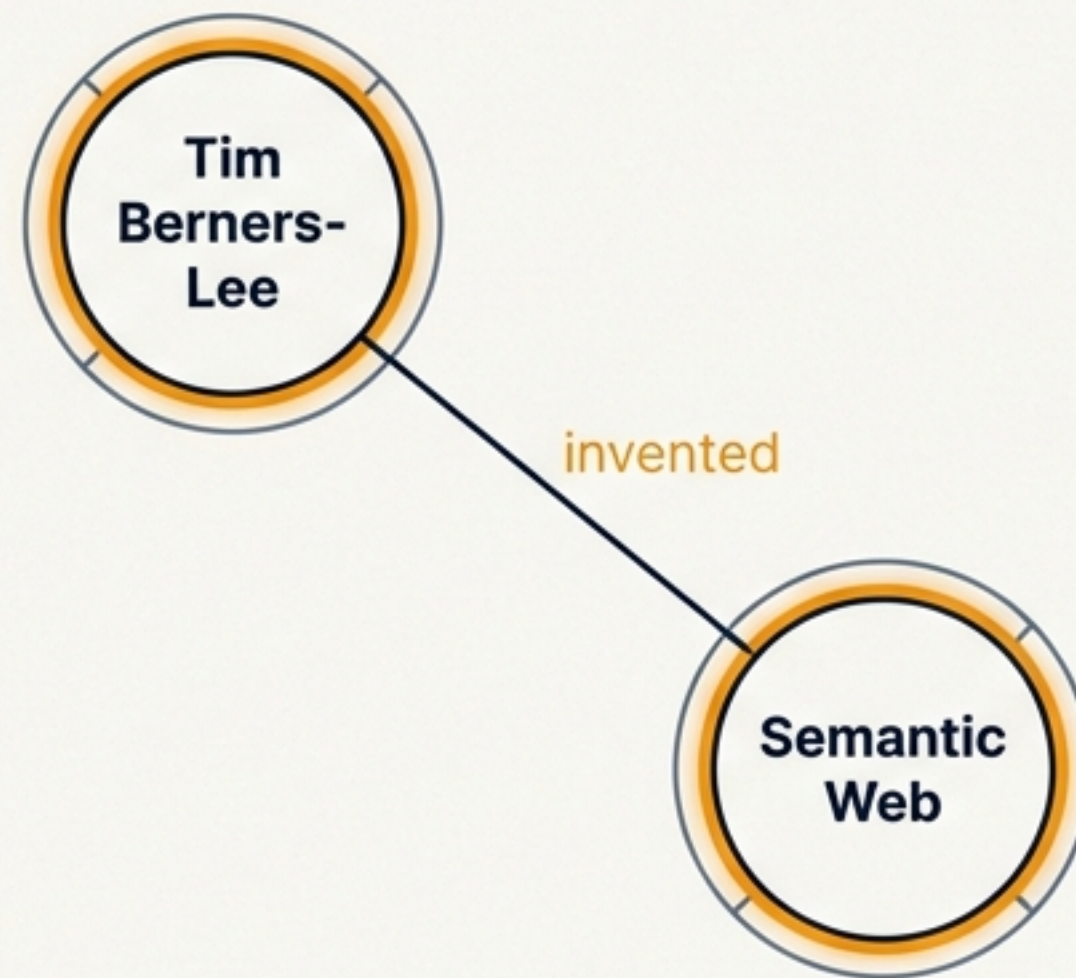
A COLLABORATION BETWEEN DINIS CRUZ AND CHATGPT DEEP RESEARCH

The Interpretability Bottleneck

The Status Quo: Vector Space



The Solution: Knowledge Graph



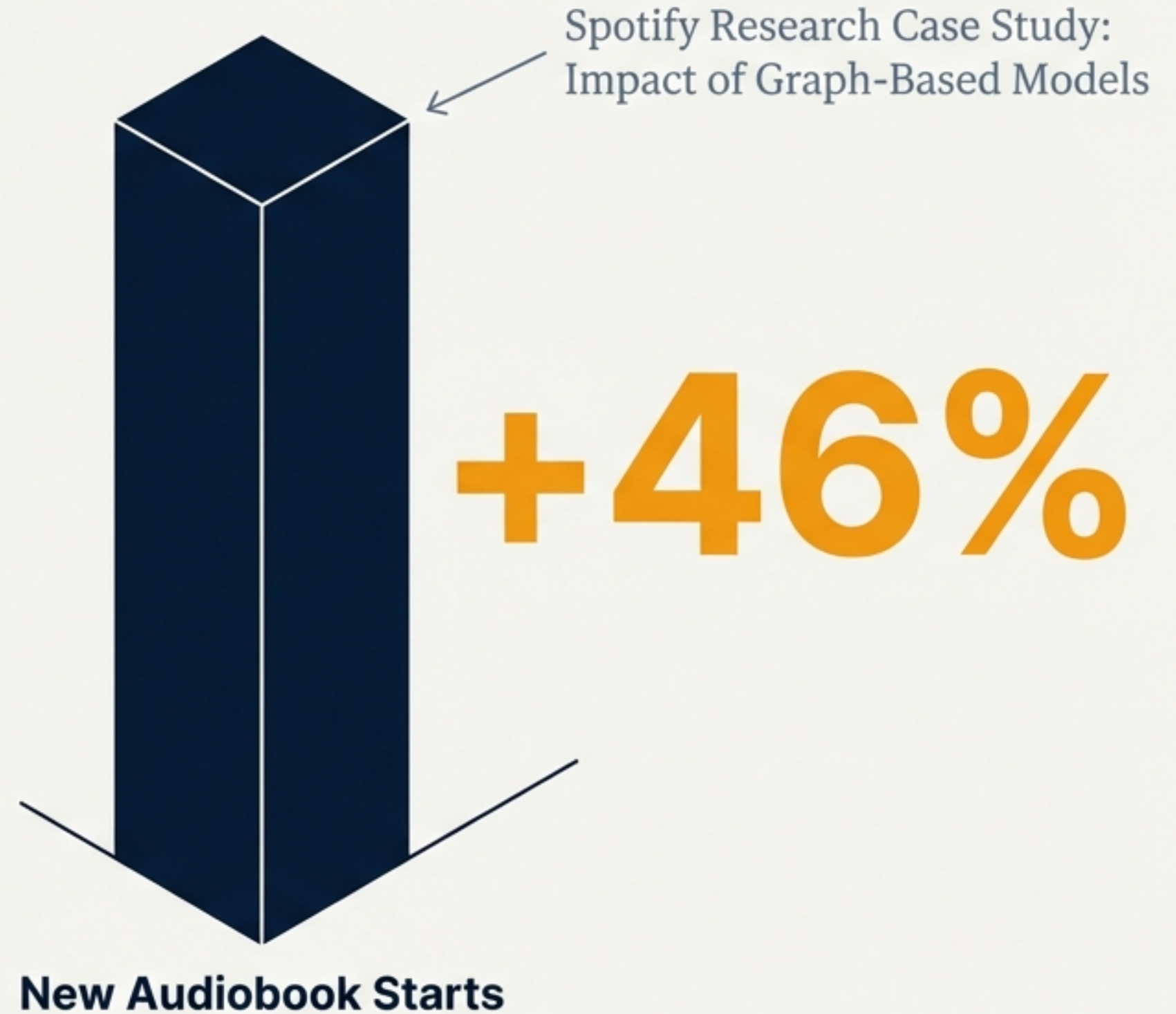
Vector-based approaches capture proximity. Graph-based approaches capture reasoning.

Modern semantic search relies on vector databases—“Black Boxes” that approximate meaning through high-dimensional mathematics. While powerful for similarity, they fail to explain *why* a recommendation was made, creating a trust gap.

Structure Facilitates Understanding

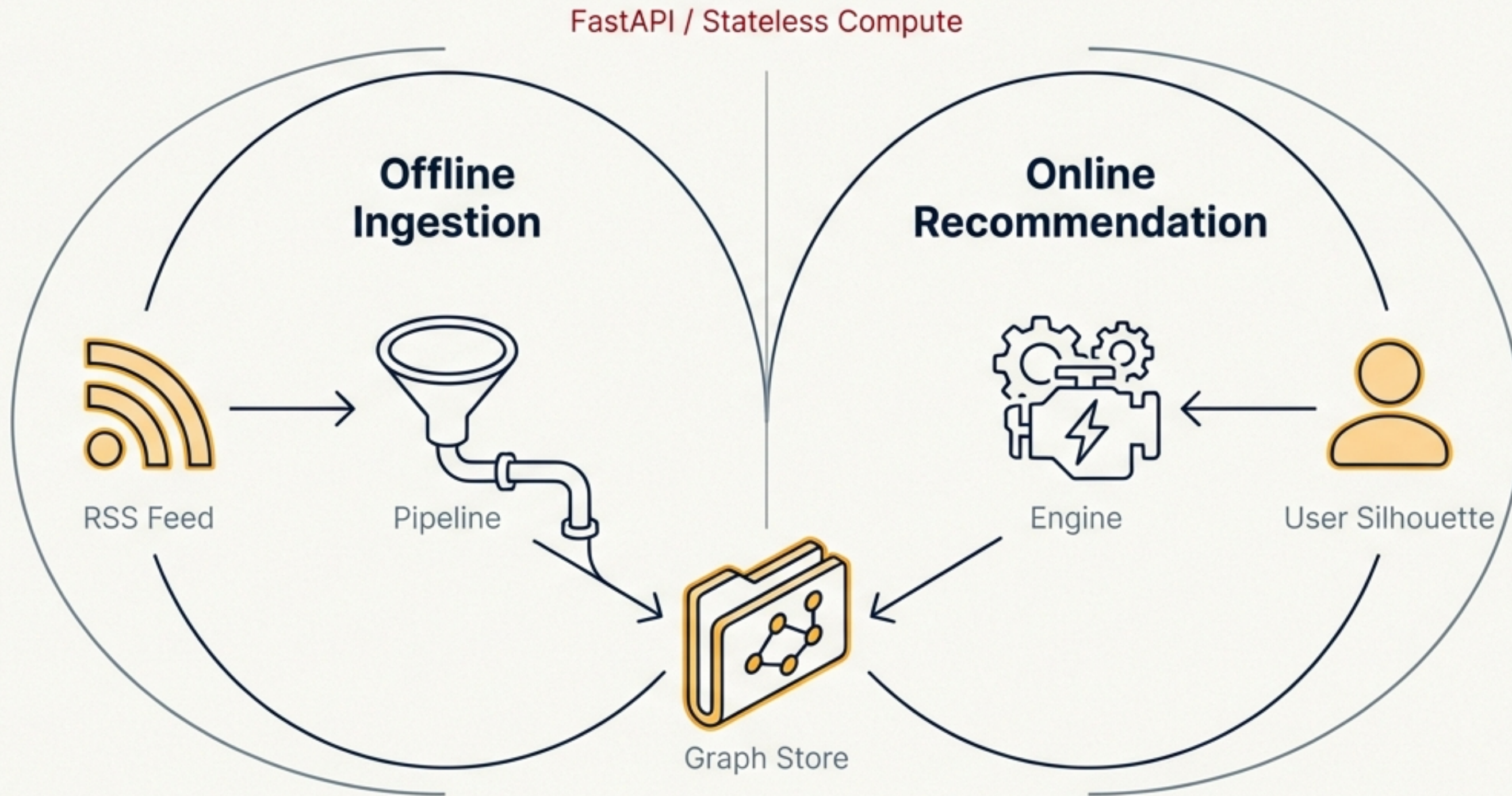
By mapping podcast content as a Knowledge Graph, we align data storage with human reasoning. This prioritises three non-negotiables:

1. **Provenance:** Knowing the source.
2. **Determinism:** Consistent outputs.
3. **Explainability:** Transparent logic.



Graph-based personalization significantly outperforms unstructured approaches at scale.

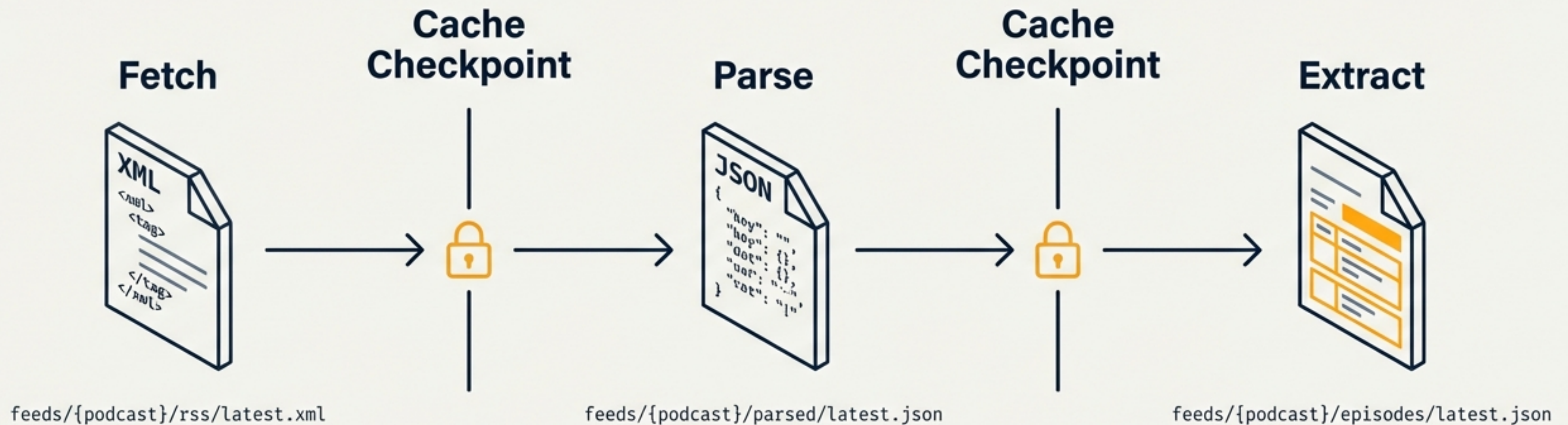
A Serverless Loop of Ingestion and Retrieval



State is persisted in Cloud Object Storage (S3/Blob). Zero database instances.

The Idempotent Ingestion Pipeline

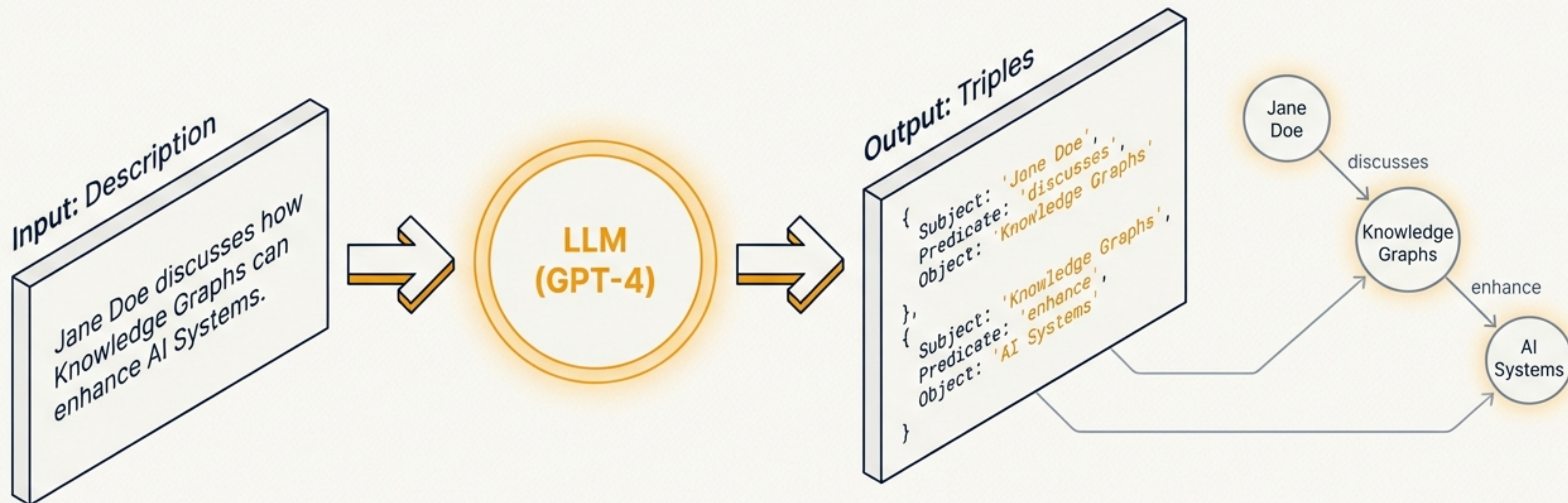
Every step is cached. Nothing is lost.



If a process fails, it resumes from the last checkpoint. If input is unchanged, processing is skipped.

Structuring the Unstructured with LLMs

We explicitly instruct the model to extract “Triples”—Entity, Relationship, Entity—rather than just summarizing.



The File System is the Database

To maximise scalability and minimise cost, the 'Graph Store' is a structured set of JSON files on Cloud Storage. Zero maintenance. Infinite scale.



Trust Through Provenance

In an era of AI hallucinations, traceability is safety.
Every node maintains a lineage back to its source.



Metadata Card

Source ID: Episode 124

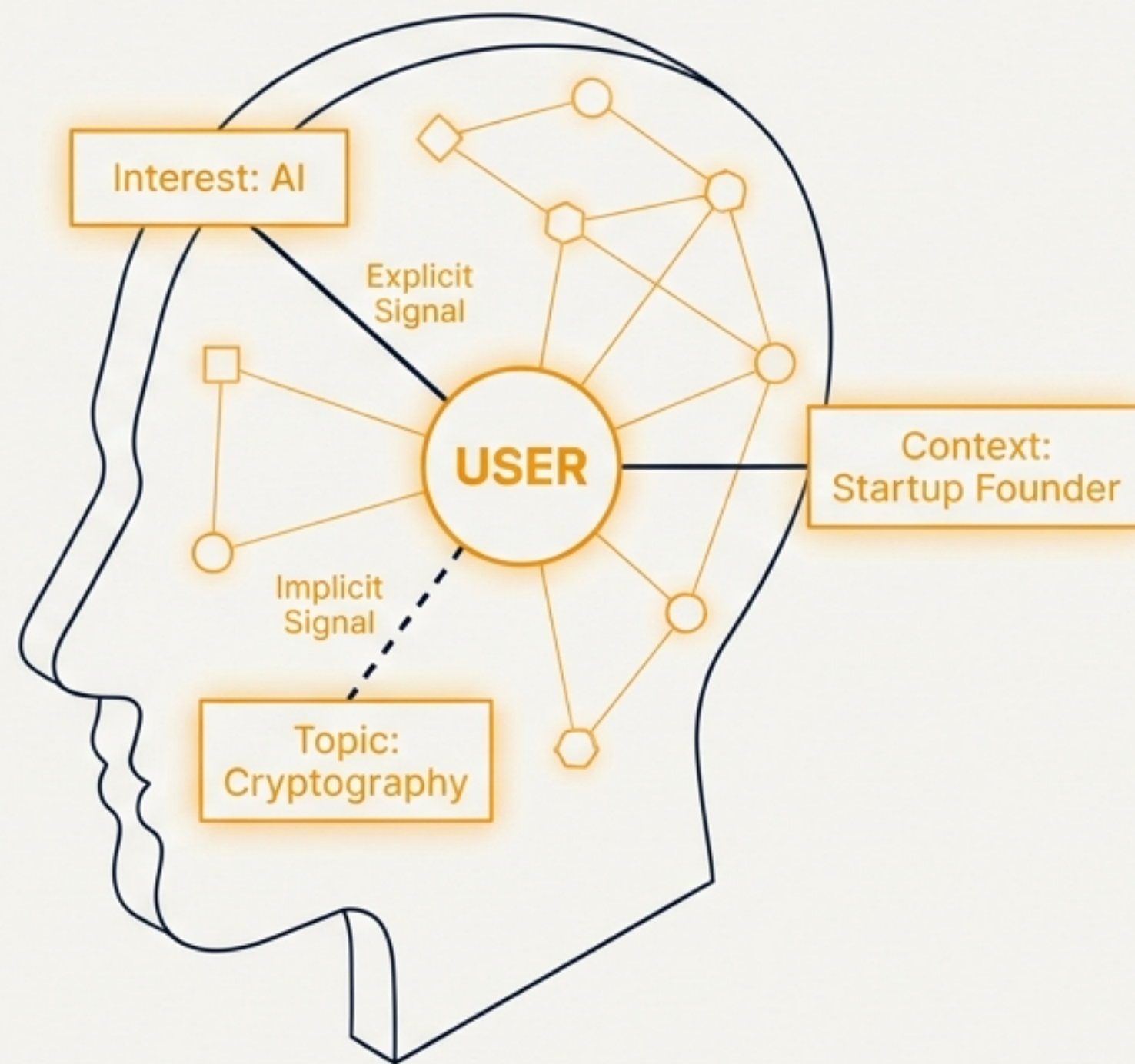
Model: GPT-4-Turbo

Date: 2026-01-20

Original Text: "This talk explores the differences between vector search and graph..."

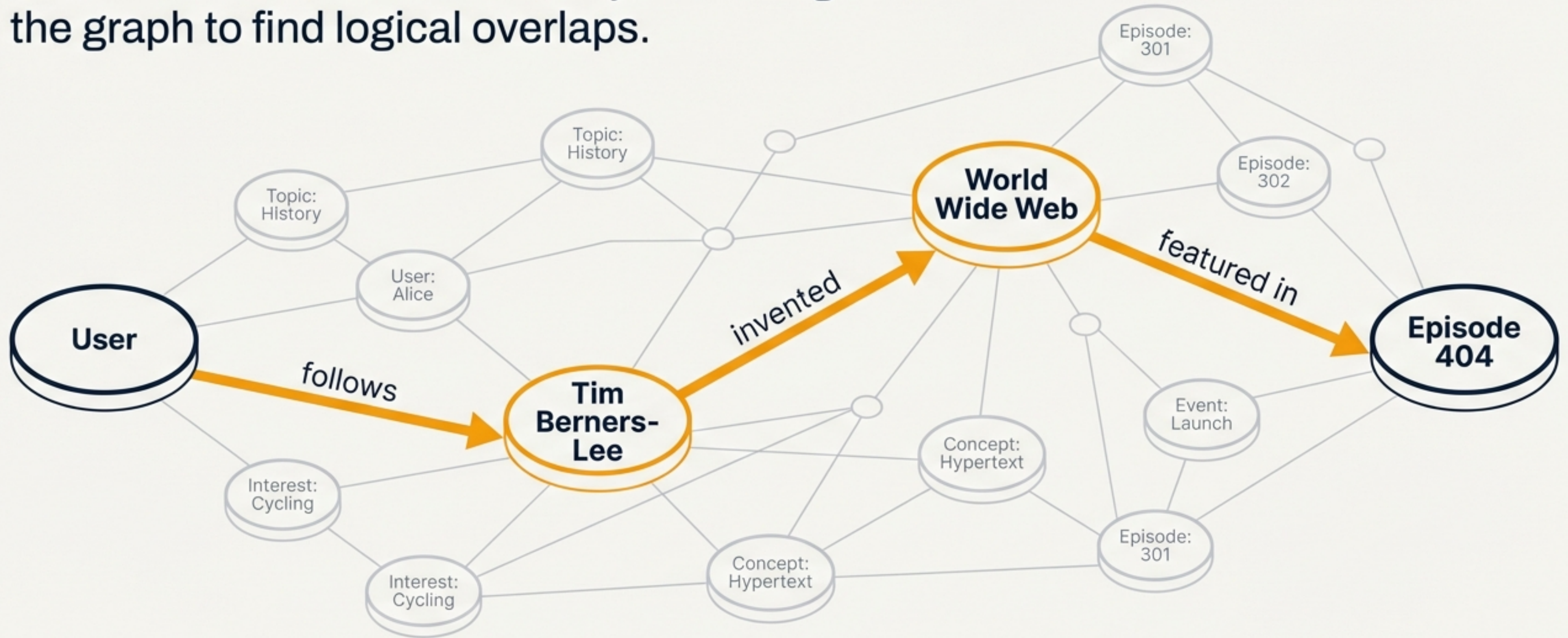
The User as a Graph

We model the user using the same semantic vocabulary as the content. Matching becomes a geometry geometry problem, not a prediction problem.



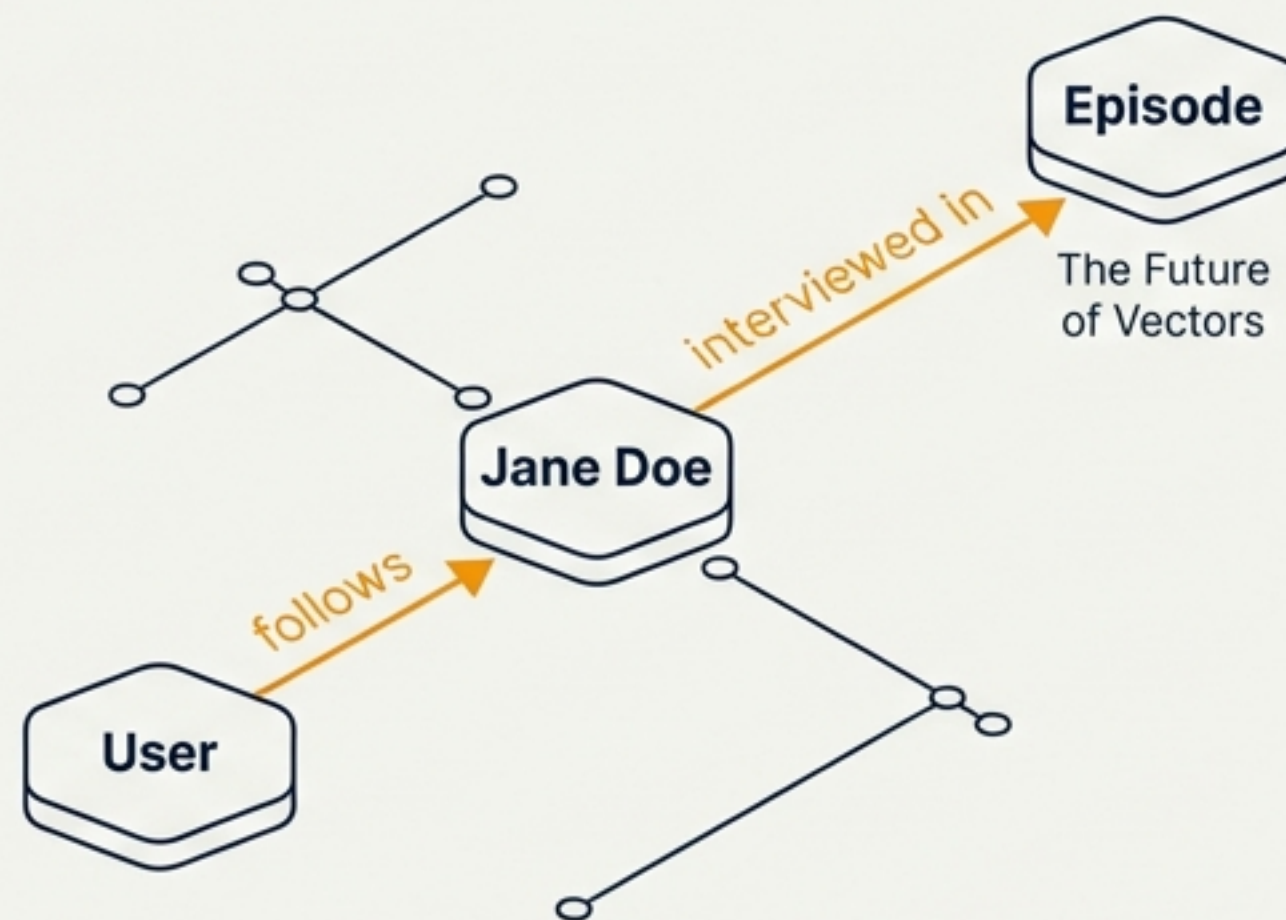
Querying for Connection

Recommendations are found by traversing the graph to find logical overlaps.



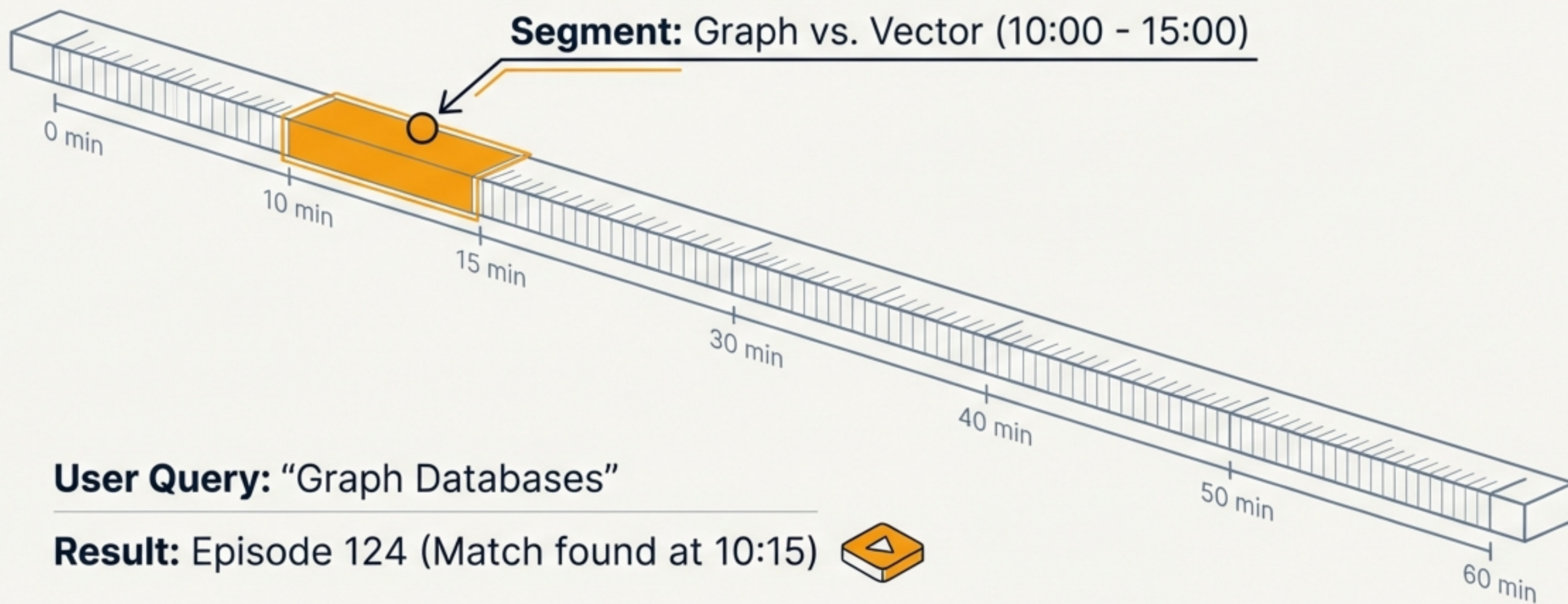
Explainability is the Interface

Transparency builds trust.



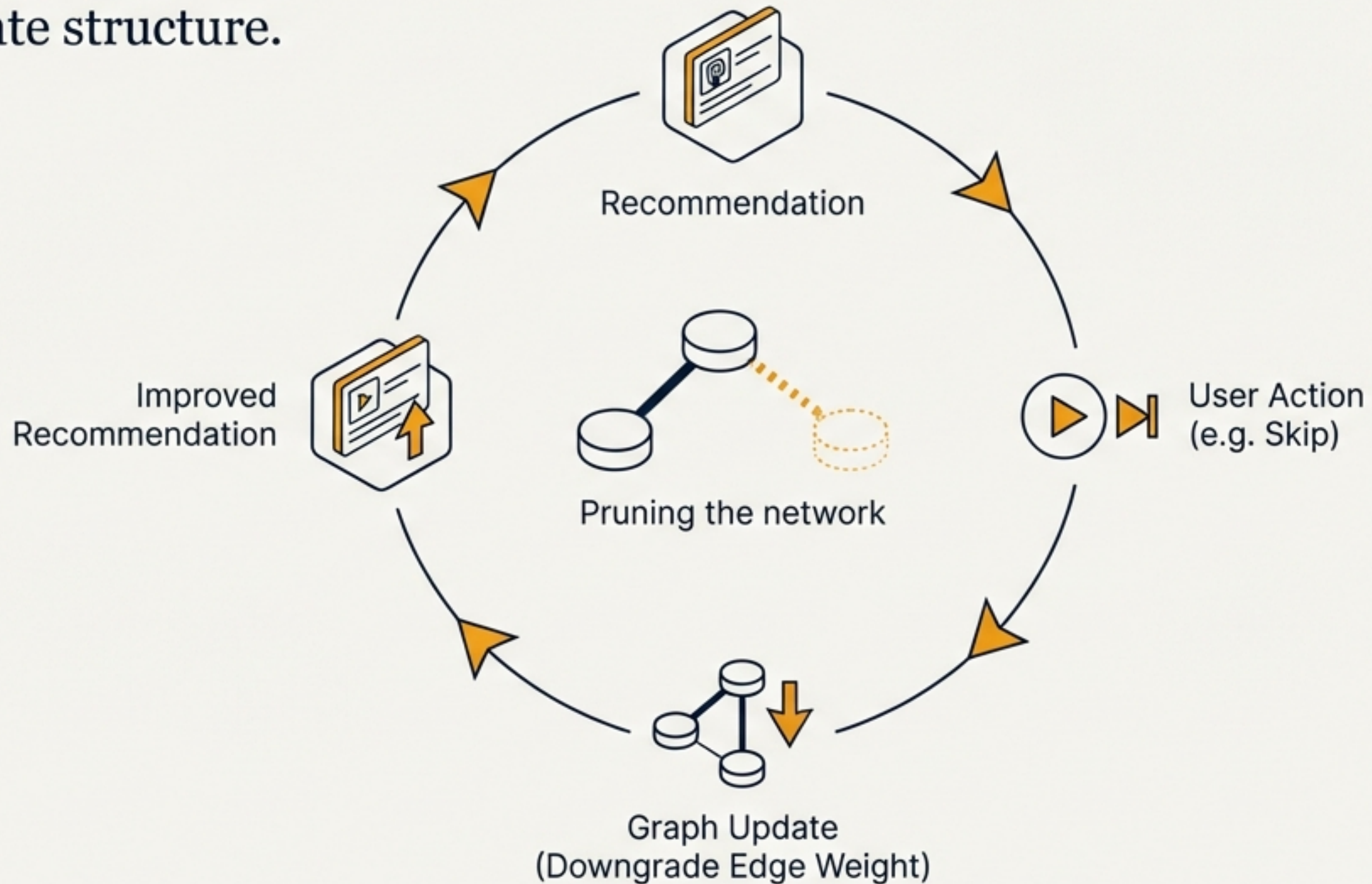
Segment-Level Discovery

Find the needle, not just the haystack. Identify specific timestamps where a topic is active.



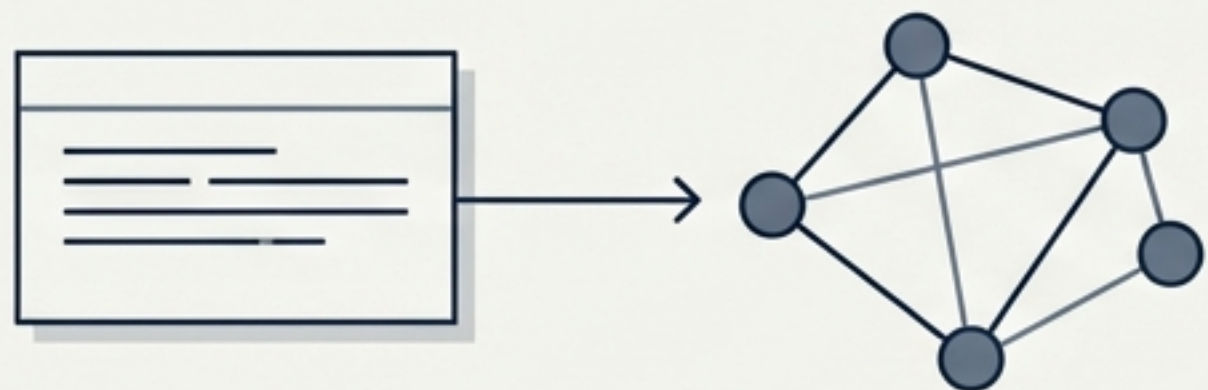
Continuous Refinement

The graph is a living organism.
User interactions provide the
signal to update structure.

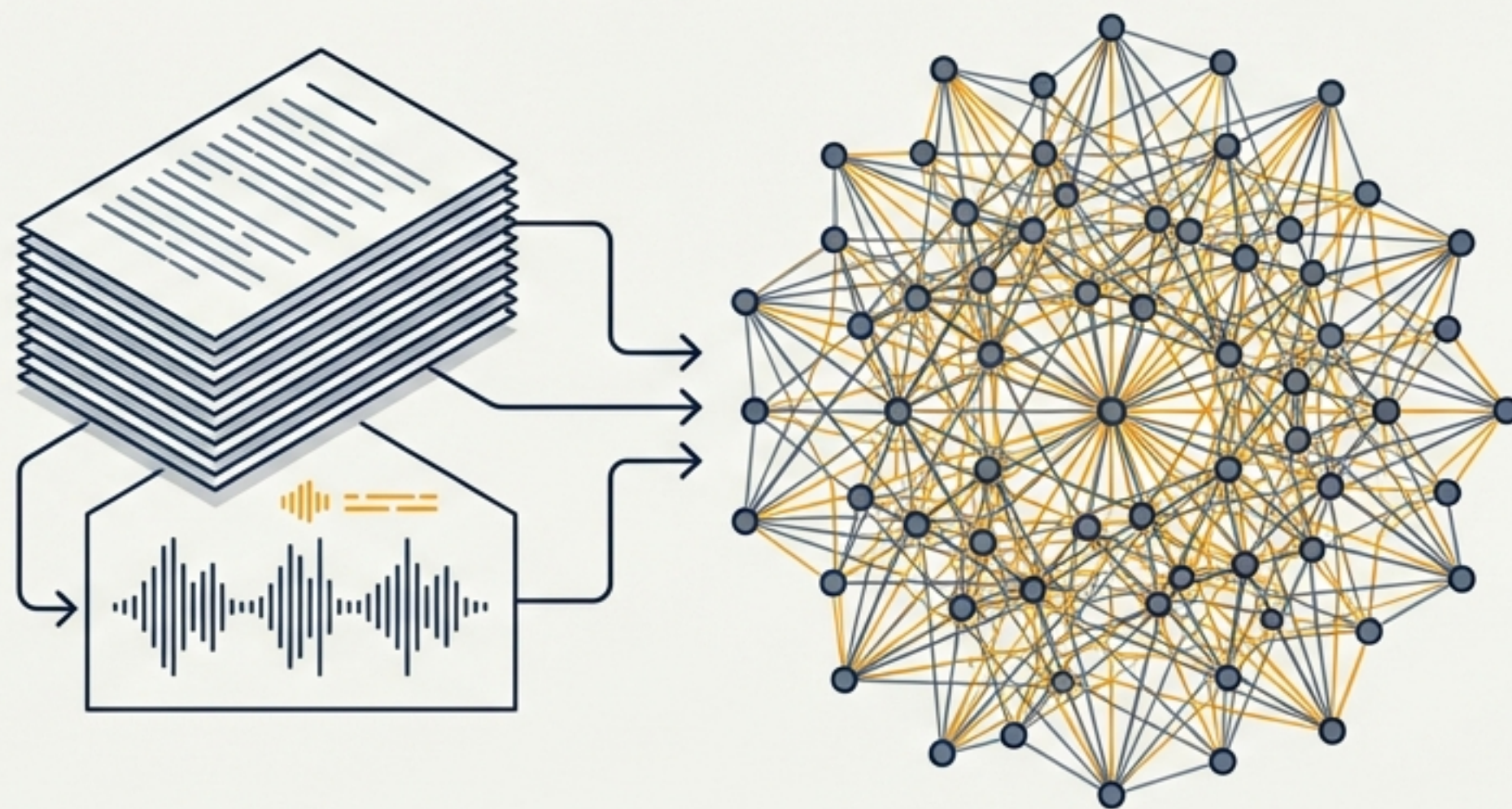


The Future: Full Audio Transcription

Phase 1: Description Analysis



Phase 2: Transcript Analysis



Moving from 100-word summaries to 5,000-word transcripts creates exponentially higher resolution for retrieval.

Turning Listening into Navigation

Structure:

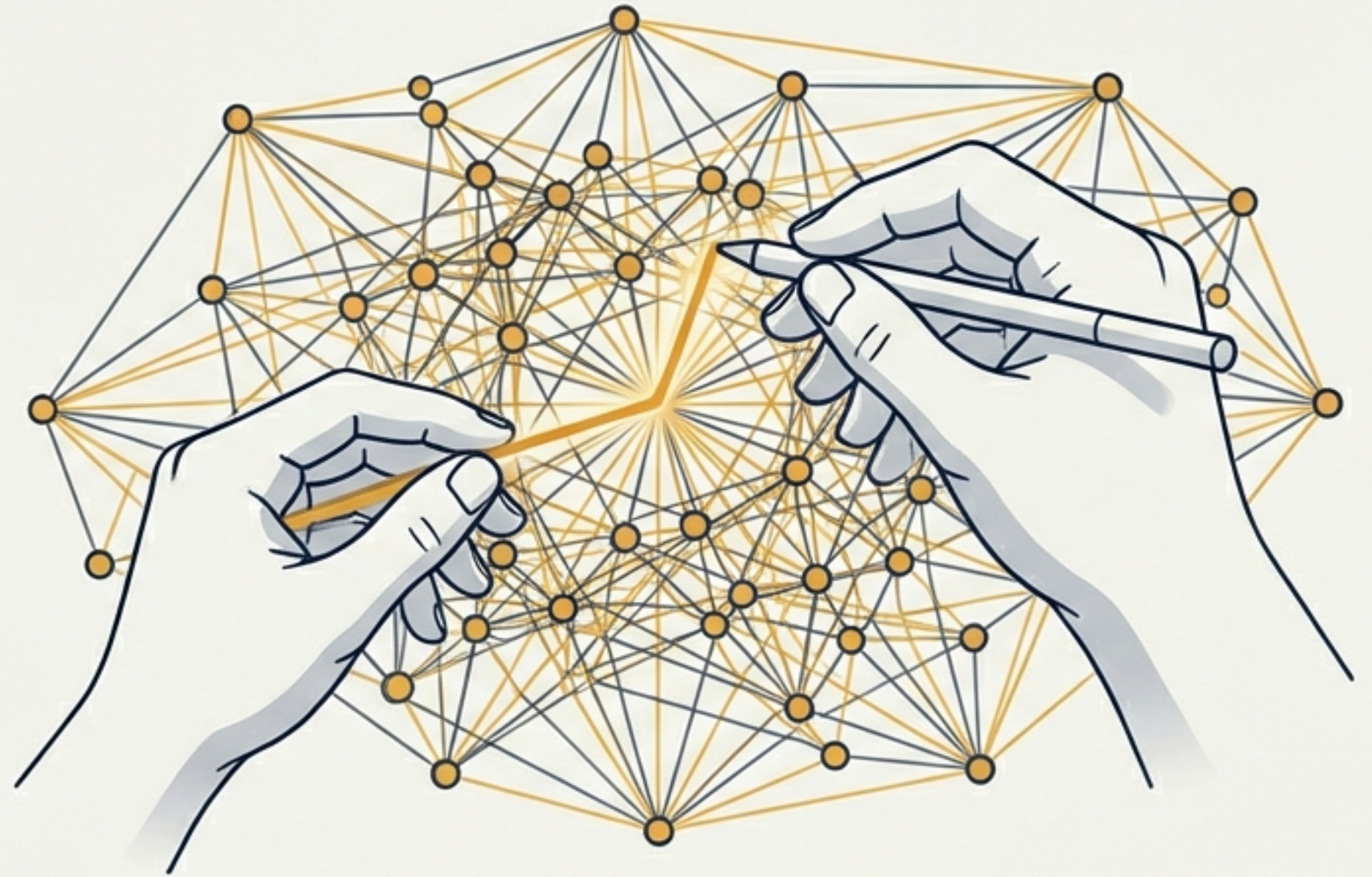
Graphs > Vectors
for explainability.

Architecture:

Serverless + Caching
= Reproducibility.

Experience:

Transparent &
Personalised.



Moving away from the 'Black Box' toward a curated, navigable knowledge experience.